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LLM Agent Based Protein Function Prediction

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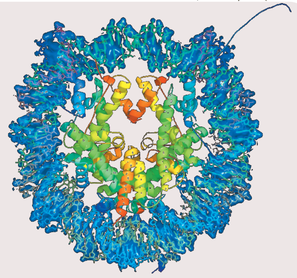
PSB 2026



Abstract

Protein function prediction methods provide (1) limited interpretability and (2) struggle to effectively integrate diverse information sources. We developed a LLM agent-based system. To address these limitations.

Our system integrates:
1. computational predictions
2. protein metadata
3. scientific literature
4. ontological information from GO
5. specialized tools

Through a multi-step reasoning process, the agent progressively refines functional predictions. Furthermore, the agent provides explanations for each prediction update.

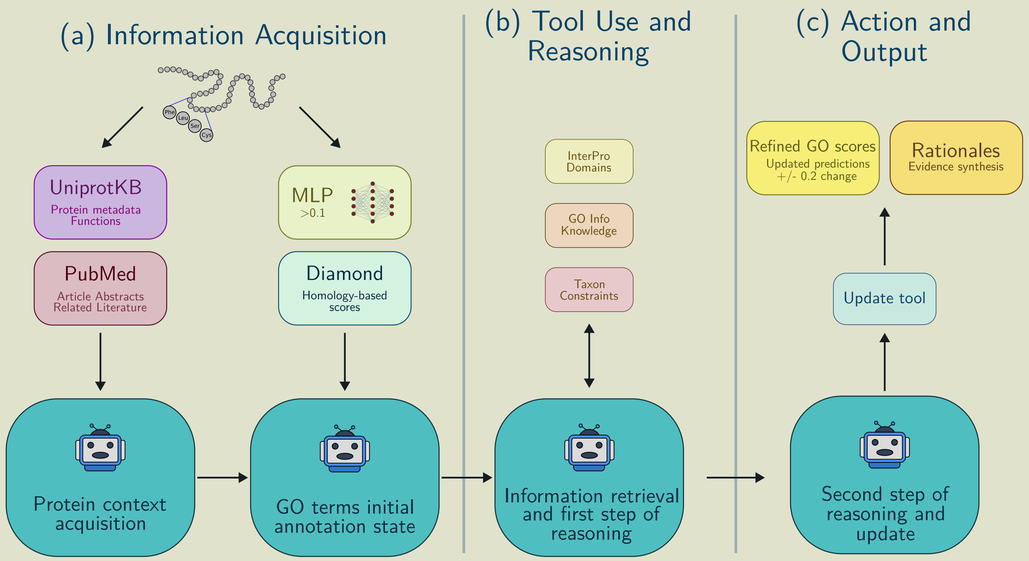
Experimental setup

We generated a time-based dataset following CAFA challenge approach. We chose the following UniprotKB releases: (a) training: v2023_05, (b) validation: v2024_06, (c) testing: v2025_03.

We used two LLMs: GPT-4.1-nano and Gemini-Flash-2.0. Both LLMs have cutoff date of June 2024.

Ontology	Terms	Proteins	Training	Validation	Testing
MFO	7,168	46,573	45,365	801	407
BPO	20,848	62,613	60,615	1,225	773
CCO	2,905	61,591	59,926	1,126	539

Methods



Results

Comparison with other methods

Method	F_{max}			S_{min}		
	MFO	BPO	CCO	MFO	BPO	CCO
Naive	0.266	0.262	0.568	12.452	30.865	9.822
DiamondScore	0.699	0.452	0.689	6.262	25.907	6.488
DeepGOPlus	0.695	0.454	0.719	6.313	25.873	6.456
SPROF-GO	0.649	0.398	0.681	7.988	27.278	7.948
DeepGraphGO	0.611	0.378	0.651	8.390	28.118	7.997
MLP	0.642	0.414	0.693	7.364	27.440	7.530
MLP+DS	0.706	**0.463**	0.718	6.177	25.495	6.365
GPT-4.1 nano	0.709	0.462	0.722	6.132	25.515	6.332
Gemini-Flash 2.0	**0.718**	0.460	**0.737**	**5.935**	**25.381**	**6.190**

Ablation study

Model	Method	F_{max}			S_{min}		
		MFO	BPO	CCO	MFO	BPO	CCO
GPT	Without Metadata	0.705	0.461	0.720	6.223	25.537	6.414
	Without Abstracts	0.708	0.461	**0.722**	**6.062**	25.530	6.351
	All info	**0.709**	**0.462**	**0.722**	6.132	**25.515**	**6.332**
Gemini	Without Metadata	0.710	0.458	0.724	6.129	25.760	6.216
	Without Abstracts	0.717	0.457	0.731	6.118	25.508	6.223
	All info	**0.718**	**0.460**	**0.737**	**5.935**	**25.381**	**6.190**

Statistical analysis

Model	Ontology	F_{max}	S_{min}
GPT	MF	0.709 ± 0.003	6.132 ± 0.046
	BP	0.462 ± 0.001	25.515 ± 0.018
	CC	0.722 ± 0.002	6.332 ± 0.053
Gemini	MF	0.718 ± 0.002	5.935 ± 0.099
	BP	0.460 ± 0.003	25.381 ± 0.111
	CC	0.737 ± 0.005	6.190 ± 0.052

Conclusions

- Our method gets better performance in threshold-dependent metrics.

- Semantic accuracy is improved across all subontologies.

- Our method provides reasoning traces that document the evidence considered during prediction refinement.

- Ablation study confirms that the additional information is used by the agent.

Future work

- Incorporation of tools that use protein sequence and provide information about:
  - structure
  - cellular location
  - binding sites

- Incorporation of expressive ontological knowledge beyond term definitions.

- Multi-agent communication between different protein agents.

Find our code at:

https://github.com/bio-ontology-research-group/go-agent